**Table 3.** Transformation modes supported by LRA.

| Mode | Matrix Formula | Mode | Matrix Formula |
|---|---|---|---|
| **affine** | $\theta = \begin{bmatrix} a & b & tx \\ c & d & ty \end{bmatrix}$ | **shear** | $\theta = \begin{bmatrix} 1 & \lambda_x & 0 \\ \lambda_y & 1 & 0 \end{bmatrix}$ |
| **translation** | $\theta = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \end{bmatrix}$ | **rotation_scale** | $\theta = \begin{bmatrix} s_x \cos\theta & -\sin\theta & 0 \\ \sin\theta & s_y \cos\theta & 0 \end{bmatrix}$ |
| **rotation** | $\theta = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \end{bmatrix}$ | **translation_scale** | $\theta = \begin{bmatrix} s_x & 0 & tx \\ 0 & s_y & ty \end{bmatrix}$ |
| **scale** | $\theta = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \end{bmatrix}$ | **rotation_translation** | $\theta = \begin{bmatrix} \cos\theta & -\sin\theta & tx \\ \sin\theta & \cos\theta & ty \end{bmatrix}$ |

**Table 4.** Hyperparameter sensitivity (SRCC avg.). Default **bolded**; ↑: peak.

| $r$ | SRCC | $\lambda$ | SRCC | $\epsilon$=$\gamma$ | SRCC | $k_1$ | SRCC | $w_{\texttt{sim}}$:$w_{\texttt{NCC}}$:$w_{\texttt{smo}}$ | SRCC | $\tau$ | F1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 0.791 | 0.1 | 0.821 | 0.0 | 0.823 | 10 | 0.819 | 1:0:0 | 0.712 | 0.3 | 0.781 |
| 8 | 0.818 | 0.3 | 0.836 | **0.1** | **0.845**↑ | 15 | 0.831 | 1:1:0 | 0.811 | 0.4 | 0.823 |
| **16** | **0.845**↑ | **0.5** | **0.845**↑ | 0.2 | 0.839 | **20** | **0.845**↑ | **1:1:1** | **0.845**↑ | **0.5** | **0.861**↑ |
| 32 | 0.841 | 0.8 | 0.831 | 0.4 | 0.831 | 25 | 0.843 | 2:1:1 | 0.840 | 0.6 | 0.849 |
| 64 | 0.833 | 1.0 | 0.819 | 0.6 | 0.812 | 30 | 0.841 | 1:2:1 | 0.837 | 0.7 | 0.832 |

### A.1 Supplementary Experimental Protocol

**Scoring and Latency.** The min-max normalisation $\phi(\cdot)$ is computed per-plane over $\mathcal{D}_B$ and *fixed* at inference to prevent distribution shift. Score weights are uniform ($w_m$=1), as no single term dominates the quality manifold; $\tau$=0.5 maximises accept/reject F1. Anchor features are ***pre-computed and cached***: with $k_1$=20, inference takes ∼5.6 ms/frame (∼180 FPS) on an RTX 4090.

**Affine Transformation.** The LRA predicts affine matrices $\Theta_i \in \mathbb{R}^{2\times 3}$. The cascaded architecture applies distinct affine transformations per feature level; the smoothness penalty regularises ***inter-level transition continuity*** rather than enforcing diffeomorphism. Table 3 enumerates the eight supported modes.

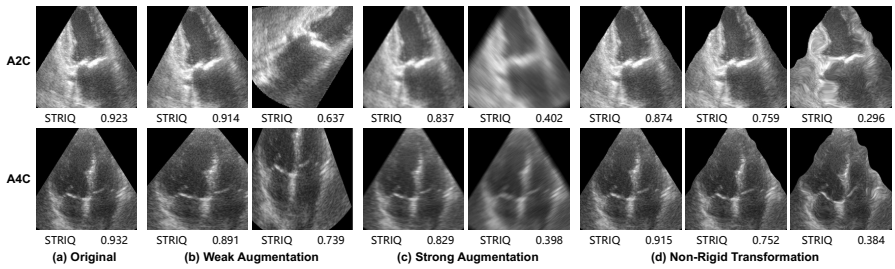


**Fig. 5.** Quality score response of STRIQ on CAMUS A2C/A4C planes under (a) original, (b) weak augmentation, (c) strong augmentation, and (d) non-rigid transformation with increasing severity (left to right within each group).

**Table 5.** *Left*: Anchor selection strategy. *Right*: $\mathcal{L}_{\texttt{orth}}$ variant (SRCC avg. on $\mathcal{D}_C$).

| Strategy | Abd. | 4CH | Kid. | Avg. |
|---|---|---|---|---|
| Random | 0.798 | 0.832 | 0.771 | 0.798 |
| K-medoids | 0.831 | 0.862 | 0.801 | 0.827 |
| K-center greedy | 0.825 | 0.855 | 0.796 | 0.821 |
| **Var.-spectrum** | **0.847** | **0.883** | **0.819** | **0.845** |

| Variant | Abd. | 4CH | Kid. | Avg. |
|---|---|---|---|---|
| $\|A_c A_{c'}^\top\|_1$ (**default**) | **0.847** | **0.883** | **0.819** | **0.845** |
| $\|A_c A_{c'}^\top\|_F$ | 0.841 | 0.876 | 0.815 | 0.839 |
| $\|\cdot\|_1$ on $A{+}B$ | 0.843 | 0.879 | 0.817 | 0.841 |
| $\|\cdot\|_F$ on $A{+}B$ | 0.838 | 0.871 | 0.811 | 0.835 |

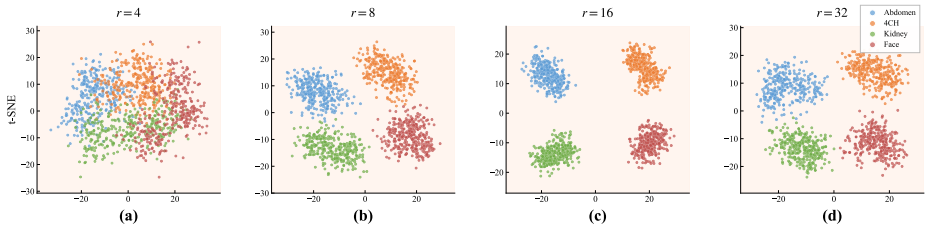


**Fig. 6.** OKS subspace embeddings under varying ranks. (a) $r$=4: overlap. (b) $r$=8: partial separation. (c) $r$=16: ***optimal compact clusters***. (d) $r$=32: mild redundancy.

### A.2 Anchor Strategy, $\mathcal{L}_{\texttt{orth}}$ Variants, and Hyperparam. Sensitivity

Table 5 (left) shows variance-spectrum selection outperforms k-medoids (+1.8%), while random selection degrades to 0.798, confirming the criticality of anchor curation. Table 5 (right) shows the default $\ell_1$ norm on $A_c$ achieves the best subspace separation (0.845), as $\ell_1$ promotes sparser cross-Gram entries than Frobenius alternatives. Extending the penalty to both projection matrices ($\|\cdot\|$ on $A{+}B$, *i.e.* $\|A_c A_{c'}^\top\|{+}\|B_c^\top B_{c'}\|$) yields no gain, indicating that penalising the down-projection cross-Gram alone suffices for subspace disentanglement. Table 4 reports sensitivity across six hyperparameters: ***OKS rank*** $r$=16 balances expressiveness and separability; ***anchor size*** $k_1$=20 suffices to capture each plane's morphological distribution, with no gain beyond this; ***threshold*** $\tau$=0.5 maximizes F1, exhibiting robustness across the $[0.4, 0.6]$ interval.

### A.3 Other Results and Subspace Analysis

**CAMUS Results.** Qualitative results in Fig. 5 show that STRIQ scores exhibit graceful decay under mild noise yet decrease proportionally to structural loss under severe deformation. This confirms that subspace-guided registrability ***captures fundamental ultrasound geometry***, effectively accommodating diverse cardiac morphologies within the jointly trained framework.

**OKS Subspace Separation.** The impact of OKS rank on feature distribution is visualized via t-SNE in Fig. 6. While a low rank ($r$=4) results in significant inter-plane entanglement, $r$=16 yields ***distinct, compact clusters*** that align with the observed SRCC peak. Increasing the rank to $r$=32 introduces marginal redundancy without further enhancing separation, reinforcing $r$=16 as the optimal balance between discriminability and manifold compactness.